PANJA BUNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd.underthePanjabUniversityActVIIof1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc. (Horticulture Sciences)

As per NEP 2020

2025-26

Course Outline for B.Voc. (Horticulture Sciences)
As per NEP 2020

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 101	Core Discipline	Weed Control	2	2	4	40	10	40	10	100
HSC 102	Core Discipline	Nursery Production	2	2	4	40	10	40	10	100
HSC 103	Core Discipline	Irrigation and Water Management	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 201	Core Discipline	Propagation Techniques	2	2	4	40	10	40	10	100
HSC 202	Core Discipline	Elements of Vegetable Growing	2	2	4	40	10	40	10	100
HSC 203	Core Discipline	Horticulture (Fruit Growing)	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Gardener and Nurseryman

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of the start of the 3rd Semester. Viva-Voce examination to beheld within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Horticulture Sciences = 48 + 4 #.

#Exit 1: Certificate Students can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during the first and second semester.

Semester –III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 301	Core Discipline	Insects & Pests of Horticultural & Vegetable Crops	2	2	4	40	10	40	10	100
HSC 302	Core Discipline	Landscaping & Floriculture	2	2	4	40	10	40	10	100
HSC 303	Core Discipline	Pomology-I	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC – III	Skill Enhancement Course	Common Basket			3					75
IDC III	Interdisciplinary Course	Common Basket			3					75
VAC – II	Value-Added Course	Common Basket			2					50
PW II	Project Work	Project Work-II& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 401	Core Discipline	Introductory Soil Sciences	2	2	4	40	10	40	10	100
HSC 402	Core Discipline	Introductory Seed Technology	2	2	4	40	10	40	10	100
HSC 403	Core Discipline	Pomology-II	2	2	4	40	10	40	10	100
HSC 404	Core Discipline	Diseases and Management of Horticulture crops	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
VAC – III	Value-Added Course	Common Basket			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Landscape & Floriculture Supervisor

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of the start of the 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of the 3rd semester.

**Minimum Credits required to get Diploma Course in Horticulture Sciences = 96 + 4 #.

#Exit 2: Diploma Students can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

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ESSENTIAL COMPONENTS OF COURSE OF B.VOC

B.Voc. (Horticulture Sciences)

(SEMESTERS I &II)

PROGRAMME OUTCOMES

PO1. To transfer knowledge of Agriculture/Horticulture in the field of agricultural research, especially in Horticulture including Fruits, Vegetables, Flowers, Species, Medicinal & Aromatic Plants & their management.

PO2. To develop innovative agro-techniques to enhance the production & productivity of Horticulture Crops.

PO3. To increase farmer's income through Hi-tech Horticulture.

PO4.To create job opportunities for unemployed youth through teaching, research, training, extension, etc.

PO5. To the establishment of model nurseries in rural areas for the availability of quality planting materials.

PO6. To prolong the post-harvest storage life of Horticulture commodities & increase income through value addition of the products & to reduce post-harvest losses.

PO7. To develop interpersonal skills under the students to identify, analyze & solve problems.

Semester I

<table>
<tr><th colspan="2">COURSE TITLE</th><th>WEED CONTROL</th></tr>
<tr><td>1.</td><td>Course Code</td><td>HSC 101</td></tr>
<tr><td>2.</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3.</td><td>Course Duration/ LP (Lecture) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical(1 credit = 15 hours of theory learning based on 2 hrs./week)
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4.</td><td>Credits</td><td>4 Credits [2+2]</td></tr>
<tr><td>5.</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks
i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
</td></tr>
<tr><td>6.</td><td>Job Role</td><td>Gardener & Nursery Man.</td></tr>
<tr><td>7.</td><td>Course Objectives</td><td>To acquaint the students with elementary information on basic agronomic principles and practices for horticulture crops.</td></tr>
</table>

8.	**Course Outcom**	On successful completion of the course, the students will be able to: CO1. To understand about harmful & beneficial effects of weed in horticulture. CO2. To identify different weeds, cycle of weeds & crop competition. CO3. To understand different methods of weed control and Integrated Weed Management. CO4. To identify and apply herbicides for the control of weeds in different horticulture crops CO5. To understand the approaches of weed management in different cropping systems
9.	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT– I

Definition of weeds, their characteristics & harmful effects. Classification of weeds, propagation & dissemination of weeds.

UNIT– II

Herbicide structure and properties, factors affecting the efficiency of herbicides, herbicide formulations, herbicide mixtures, Degradation of herbicides in soil and plants, weed shift in cropping systems, weed control through bioherbicides and allelochemicals, herbicide-resistant crops, herbicide rotation.

UNIT– III

Weed control/ management (Legislative, physical, chemical, mechanical, biological) in cropping systems and non–cropped situations–control of parasitic weeds – control of aquatic, biennial, and perennial weeds.

UNIT-IV

Weed control/ management in major horticultural crops. Integrated weed management

Practical work:

1. Identification of important weeds and preparation of herbarium
2. Calculation of herbicide requirements
3. Use of various types of spray pumps and nozzles
4. Preparation of spray solutions and application of herbicides

Suggested Resources for Reading

1. Ashton, F.M and Crafts, A.S. 1981. Mode of Action of Herbicides (2nd Ed). Wiley Inter-Science
2. Gupta, O.P. 2007. Weed Management – Principles and Practices. Agro bios
3. Mandal, R. C, 1990. Weed, Weedicides and Weed Control – Principles and Practices. Agro–Botanical Publ.
4. Rao, V.S. 2000. Principles of Weed Science. Oxford & IBH

Note: Latest edition of textbooks may be used.

	COURSE TITLE	NURSERY PRODUCTION
1.	**Course Code**	HSC 102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Self-Employment in the Business of Nursery.
7.	**Course Objectives**	To make the students familiar with the basic principles and procedures of nursery raising for horticulture crops.

8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Familiarization with principles & practices of propagation & nursery management for Horticultural Crops. CO2. Study of introduction to propagation, cellular basis for propagation, sexual propagation, apomixes, polyembryony, chimeras. CO3. Principal factors influencing seed germination of horticulture crops, dormancy, hormonal regulation of germination & seedling growth. CO4. Knowledge of nursery management, nursery establishment & nursery rules & regulations. CO5. Imparting knowledge about varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients and plant protection, hardening, packaging, storage, transport, and marketing of vegetables.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT– I

Scope and importance of vegetable and flower nursery planning I - location, soil, topography, layout, paths, roads, nursery beds, office, store, potting yard planning II - water, containers, electricity, fencing, shelter belts – Propagation structures - greenhouses, Mist chambers, poly houses – growing media - selection of media, quality of media, soil, sand, peat, leaf mold, saw dust, sphagnum moss, coco peat.

UNIT– II

Propagation – seed germination – dormancy-methods of breaking dormancy – pre-sowing seed treatment. methods of sowing – irrigation systems- nutrient management – water soluble and liquid fertilizers

UNIT– III

Varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients, packaging, storage, transport, and marketing of tomato, brinjal, chili, capsicum, cabbage, cauliflower, carrot, radish.

UNIT-IV

Varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients, packaging,storage, transport, and marketing of mango, apple, pear, peach, and guava.

Practical work:

1. Features of nursery-factors to be considered for setting up of nursery
2. Studying the problems of nursery business
3. Visit to commercial vegetables, flower crops nursery
4. Seed sowing-vegetable crops-annual flower crops
5. Application of growth regulators for nursery
6. Irrigation and nutrient management-liquid fertilization
7. Care of nursery plants (shifting of plants, weeding, shade provision, watering, hardening).

Suggested Resources for Reading

1. Nanda, K.K. and V.K. Kochhar; 1991. Vegetative propagation and management. Indus Publishing Company, New Delhi.
2. Sadhu, N.K. 1989. Plant propagation, Wiley Eastern Ltd., New Delhi.
3. Agarwal, R.L. 2005. Seed Technology. Oxford and IBH Publishing Co. New Delhi.

Note: Latest edition of textbooks may be used.

	COURSE TITLE	IRRIGATION AND WATER MANAGEMENT
1.	**Course Code**	HSC 103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practicals for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	See table and instructions below

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Technician in Irrigation Department.
7.	**Course Objectives**	To acquaint students with a soil-water relationship in the context to agriculture/ horticulture crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart knowledge and understanding of fundamental concepts and techniques of maintaining and enhancing water management. CO2. Develop the ability to skill the practices and their scope to improve inthe water management. CO3.Impart knowledge and understanding of sources, modes, and systems of irrigation CO4. Familiarization with different methods of water conservation.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT– I

Irrigation and its importance, Importance of water to plants, Time of irrigation. Principal of irrigation (soil moisture, root zone depth, water holding parameters, effective irrigation, daily water use) Basis for scheduling irrigation to crops: plant basis approach, climatologically approach and soil moisture depletion approach.

UNIT– II

Sources and modes of irrigation (canals, wells, tube wells, tank irrigation), irrigation system of Punjab, Flood irrigation, furrow irrigation, sub surface irrigation, overhead irrigation, manual irrigation, localized irrigation and sprinkler irrigation. Methods of measurement of irrigation, water requirement of different crops.

UNIT– III

Water logging, different types of water movement in soil. Principal of drainage: surface drainage and sub surface drainage. Kind of sub surface drains. Crop response to quality of irrigation water. Irrigation management and problem, Soil irrigation for frost protection.

UNIT-IV

Erosion Problems in Punjab and India, type of Erosion: water Erosion, Sheet and Rill Erosion, Scalding Erosion, Gully erosion, Tunnel Erosion, Stream Bank Erosion, Mass Erosion and Wind Erosion, Causes and Effects of Erosion, Factors responsible for Water and Wind Erosion. Form of wind and water Erosion. Different methods of water conservation.

Practical work:

1. Determination of soil moisture by different methods. Need for irrigation.
2. Measurement of irrigation water.
3. Preparation of land for different methods of irrigation.
4. Visit to area affected by erosion.
5. Visit to soil conservation research and demonstration center.

Suggested Resources for Reading

1. Iraeslon, Irrigation Principles, John Willey & Sons New Delhi 1996.
2. Gandhi R T., Gupta, P.C., Joseph, A.P. and Rage, N. I, Handbook of Irrigation WaterManagement.
3. Mickael, A.M., Irrigation Theory and Practices.

Note: Latest edition of text books may be used.

Semester II

	COURSE TITLE	PROPAGATION TECHNIQUES
1.	**Course Code**	HSC-201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	See below
6.	**Job Role**	Gardeners
7.	**Course Objectives**	To get basic knowledge of various propagation techniques to develop practical skills.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Familiarization with principals & practices of propagation & nursery management for Horticultural Crops. CO2. Study of introduction to propagation, cellular basis for propagation, sexual propagation, apomixes, polyembryony, chimeras. CO3. Principal factors influencing seed germination of horticulture crops, dormancy, hormonal regulation of germination & seedling growth.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

10. Course Content

UNIT– I

Propagation: Need and Potentialities, Sexual and Asexual methods of propagation and their advantages and disadvantages Seed Dormancy: Causes (Internal and external factors) Seed Dormancy: Methods of breaking Dormancy

UNIT– II

Cutting Techniques: Stem Cutting, Leaf Cutting, root cutting Layering Techniques: tip layering, simple layering, compound layering, mound layering, air layering

UNIT– III

Grafting techniques, necessity of Grafting in plants and backstop. Type of Grafting: cleft Grafting, Bark Grafting, Side-veneer Grafting, Splice Grafting, Whip and Tongue Grafting, Saddle Grafting, Bridge Grafting.

UNIT-IV

Methods of Vegetative Propagation: Budding techniques: T or Shield Budding, Chip Budding, Inverted T-Budding, I-Budding, Flute Budding, Patch Budding. Advantages of Budding and safeguard during budding.

Practical work:

1. Horticultural tools and implements used for various operations.
2. Preparation of pot mixture, potting and repotting.
3. Preparation of growth regulators and method of application in horticultural crops.
4. Demonstration of propagation through layering and cutting.
5. Demonstration of propagation through budding.
6. Demonstration of propagation through grafting and top working.
7. Propagation through specialized plant parts.
8. Methods of manuring and irrigation.

Suggested Resources for Reading

1. Bose, T.K. Sanyal, D and Sandhu, M.L. (1998) Propagation of Horticultural crops. Naya Prakash Publishers, Kolkata
2. Sadhu, N.K. 1989. Plant propagation, Wiley Eastern Ltd., New Delhi.
3. Nanda K. K., Kochhar V.K. (1985). Vegetative Propagation of Plants. Kalyani Publisher- New Delhi-Ludhiana.
4. Hartmann, H.T., Kester D.E., Davis, F.T and R.L Geneve (2010). Plant Propagation: Principles and Practices (8th Edition)
5. Grewal, H.S. 1999. Propagation of ornamental plants, Kalyani Publishers, New Delhi
6. Hartmann, H.T. and D.E. Kester; 1989. Plant propagation-Principles and practices. Prentice – Hallof India, New Delhi.
7. Nanda, K.K. and V.K. Kochhar; 1991. Vegetative propagation and management. Indus Publishing Company, New Delhi.

Note: Latest edition of text books may be used.

	COURSE TITLE	**: ELEMENTS OF VEGETABLE GROWING**
1.	**Course Code**	HSC-202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP** practical **(Lectures)** hrs./week) **(Practical)** hrs./week)	30 hours of lectures for Theory + 60 hours of (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Supervisor in Vegetable Industry.
7.	**Course Objectives**	To get cultivations and other agronomic practices for raising vegetable crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. To educate production technology of seasonal vegetables. CO2. To imparting knowledge about botany & taxonomy, climate & soil requirements, commercial varieties/hybrids, sowing/planting time & methods, seed rate & seed treatment, nutritional & irrigation requirements, intercultural operations, weed control, mulching, physiological disorders, harvesting, post harvest management, plant protection measures, & seed production of vegetables.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10. Course Content

UNIT– I

Scope, Importance and Types of vegetable growing, importance of kitchen garden. Classification of vegetable crops.

UNIT– II

Cultivation practices including Soil and Climatic requirements, varieties, seed rate, time of sowing, spacing, fertilizers and manures, irrigation, hoeing and yield potential of Cole Crops, Root Crops, Bulb Crops and their harvesting.

UNIT– III

Cultivation practices including Soil and Climatic requirements varieties, seed rate, time of sowing, spacing,fertilizers and manures, irrigation, hoeing, and yield potential of Peas, Okra, eggplant and their harvesting.

UNIT-IV

Storage and seed production of the Potato, Cole Crops, Root Crops, Bulb Crops Peas and Solanaceous fruits.

Practical work:

1. Identification of various vegetable crops, varieties and seeds.
2. Preparation of the field for sowing/ transplanting of various crops.
3. Application of manure and fertilizers, watering, hoeing, other cultural operations.
4. Prepare calendar of operation for various vegetable growing in Punjab.

Suggested Resources for Reading

1. Dhaliwal M.S., Handbook of vegetable crops, Kalyani Publishers, Ludhiana, 2008.
2. Package of Practices for Vegetable crops, P.A.U. Publications Ludhiana, Corresponding year.
3. Das, P.C., Vegetable crops of India, Kalyani Publishers, Ludhiana, 1993.
4. Chauhan, D.V., Vegetable production in India, S. Ram Prasad and Sons, Agra, 1993

Note: Latest edition of textbooks may be used.

	COURSE TITLE	**HORTICULTURE (FRUIT GROWING)**
1.	**Course Code**	HSC-203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks**)** 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	**External Theory**: 40 | **Internal Theory**: 10 | **External Practical**: 40 | **Internal Practical**: 10 **Theory Paper:** **(4x 2 Marks) = 08 Marks** **(4x 8 Marks) = 32 Marks** i) The exam of this course will be of 3 hours duration. ii) There will be no objective-type questions. iii) Students are required to know about the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees. **Instructions for paper setters and candidates:** i) The syllabus of this paper has been divided into FOUR units. ii) All questions carry equal marks. iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each. iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
6.	**Job Role**	Horticulture Laboratories & Field Technician.
7.	**Course Objectives**	To learn agronomic and other management practices in fruit growing. The Student will also learn fruit preservation method & their scope.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. To understand the principle, and theoretical aspects & developing skills in fruit cultivation or horticultural crops. CO2.To imparts basic knowledge about the importance & management of fruits grown in India. CO3. Study of commercial varieties of regional, national & international importance, eco-physiological requirements, recent trends in propagation, rootstock influence, planting system, cropping systems, cropping system, root zone & canopy management, nutrient management, water management, fruit set & development, etc. CO4. To execute quality improvement by management practices.

9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10 Course Content

UNIT– I

Fruits and their importance, Selection of site and soils for planting an orchard. Orchard soil management: clear culture, mulching, sod, sod mulch, intercropping, mixed cropping, multitier cropping, organic farming.

UNIT– II

Importance and scope of fruit growing in Punjab such as increase production, increase irrigation facilities, increase area under dry land, increase urbanization, increase transport facilities, increase cold storage facilities and pre cooling center, development new techniques, availability of loan, government incentive, export of fruits.

UNIT– III

History and development of fruit preservation and its scope. Canning of fruit (Gooseberry, Blueberry, Blackberry, Apple, Pear) Bottling of fruits (Banana jam, Lemon curd, Spiced citrus fruit, pickled Pears, Blueberry jam).

UNIT-IV

Different methods (canning, cooling, pasteurization, heating, boiling and adding sugar, salt and additives) of preservation and preparing jams, jellies and marmalades.

Practical work:

1. Identification of various vegetable crops, varieties and seeds. Preparation of the field for sowing/ transplanting of various crops.
2. Application of manure and fertilizers, watering, hoeing, other cultural operations. Prepare calendar of operation for various vegetable growing in Punjab.
3. Practice in propagation, methods, layout of orchard, transplanting, manuring, training and pruning of fruit plants.
4. Studies of cold storage and quick-freezing practices and refrigerated transport methods.

Suggested Resources for Reading

1. Singh, R., Fruits, National Book Trust of India, New Delhi, 1969.
2. Package of Practices for Cultivation of Fruits, P.A.U. Publications Ludhiana, Corresponding year.
3. Hayes, W.B., Fruit growing in India, Kitabistan, Allahabad, 1945.
4. Lal, G., Siddappa, S. And Tandon, G.L., Preservation of fruits and vegetables, Indian Council of Agricultural Research, New Delhi, 2009.

Note: Latest edition of text books may be used.

	COURSE TITLE	**INSECTS AND PESTS OF HORTICULTURAL AND VEGETABLE CROPS**
1.	**Course Code**	HSC 301
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP** **(Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week)(1 Semester = 15 weeks**)** 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5	**Max. Marks, Time and Instructions**	

External Theory	**Internal Theory**	**External Practical**	**Internal Practical**
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Supervisor in plant protection clinics
7.	**Course Objectives**	To make the students familiar with various insects -pests of horticulture and vegetables crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1: Familiarized with identification of different insect pests of horticulture, ornamentals, vegetables, and stored grains at the field level. CO2: To be able to understand the biology, diversity, distribution of insects, and their relationships to crops and the environmental conditions of a particular area. CO3: To understand identification of the nature of damage and symptoms caused by the pest so suitable technique of pest management can be applied for effective control. CO 4: Management of crop pests through the Integrated Pest Management approach without side effects on plant, animal, and environmental health.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10 Course Content

UNIT– I

Pest: definition and its categories, losses from pests to agricultural crops and their products. Natural Control and factors causing outbreak of pests of agricultural importance.

UNIT– II

Principle and methods of pest control in reference to IPM-its components viz. physical, legal, cultural, biological and chemical. Concept of economic injury and economic thresh hold level.

UNIT– III

Distribution, host range, damage and control of insect-pest of temperate fruits (apple root borer, apple stem bore; peach stem borer, peach fruit fly, cherry stem borer)

Distribution, host range, damage and control of insect pest of sub tropical fruits (citrus caterpillar, citrus psylla, litchi bug, bark eating caterpillar of loquat.)

UNIT-IV

Distribution, host range, damage and control of insect-pests of winter vegetables (potato tuber moth, potato aphid; onion maggots; pea leaf minor, turmeric leaf roller)

Distribution, host range, damage and control of insect-pests of summer vegetables (melon fruit fly, red pumpkin beetle: sweet potato weevil, chillies thrips, aphids)

Practical work:

1) Identification of various above-mentioned insect-pests, damaging stages, and their control measures
2) Preparation of an insect-pest box of various above-mentioned insect-pests

Suggested Resources for Reading

1) Srivastva, K.P., Text book of Applied Entomology, Kalyani Publishers, Ludhiana, 2009
2) Atwal, A.S., and Dhaliwal, G.S., Agricultural pest of south Asia and their management, Kalyani Publishers Ludhiana, 2005
3) Singh, R. Fruits, National Book Trust of India, New Delhi, 1969
4) Das, P.C., Vegetable crops of India, Kalyani Publishers, Ludhiana,1993
5) Package of Practices for Cultivation of Vegetable Crops, P.A.U. Publications Ludhiana, Corresponding year
6) Package of Practices for Cultivation of Fruits Crops, P.A.U. Publications Ludhiana, Corresponding year

Note: Latest edition of text books may be used.

	COURSE TITLE	**LANDSCAPING AND FLORICULTURE**
1.	**Course Code**	HSC 302
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** on 2 hrs./week)**(Practical)** hrs./week)	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based (1 credit = 30 hours of practical based on 4 (1 Semester = 15 weeks**)** 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Landscaping and Floriculture Supervisor
7.	**Course Objectives**	To get basic principles in landscaping and knowledge for the cultivation of ornamental plants,shrubs and other floriculture plants.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1 The student will know fundamental gardening principles, which form the basis for learning landscape architecture. CO2- The students will be able to identify different soft and hard landscape elements. CO3- The students will become capable of designing a garden project and preparing the cost economics. CO4- The students prepare garden layouts to a specific scale for big commercial projects
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10. Course Content

UNIT– I

Principles of landscaping and floriculture, designs and styles of gardening, layout of home-gardens and indoor-scaping, layout of different styles of formal and informal gardens

UNIT– II

Utilization of important Trees, Shrubs, Climbers, Ground covers, Potted and Shade loving plants for gardening and landscaping

UNIT– III

Lawn management, selection of grass, soil preparation, planting of grass, diseases and other problems Importance of lawn in urban and rural areas and effect on trade

UNIT-IV

Commercial floriculture cultivation practices and propagation method of Chrysanthemum, Gladioli, Rose and Marigold each including detailed study of climate, soil, timing production, land preparation, different varieties and fertilization

Practical work:

1. Identification and maintenance of seasonal flowers, trees, shrubs, creepers & indoor plants
2. Raising of seedlings
3. Identification of implements and tools of landscaping and floriculture
4. Preparation of herbarium
5. Layout of home-gardens and lawns

Suggested Resources for Reading

1. Salaria, A.S. 1999. Horticulure at Glance.M/s Jain Brothers, New Delhi
2. Des Raj 2002. Floriculture and Landscaping. 1st Edition, Kalyani Publishers, Ludhiana, India
3. Des Raj 2006. Objectives Floriculture and Landscaping, 1st Edition, Kalyani Publishers, Ludhiana,
4. India
5. Arora, J.S. 1998. Introductory Ornamental Horticulture, Kalyani Publishers, Ludhiana
6. Gopalaswami, I. 2009. Complete Gardening in India - ICAR New Delhi
7. Randhawa, M.S. 2008. Beautiful trees and gardens, National book trust-New Delhi

Teachers and students to surf concerned/internet sites including Wikipedia for subject/topic updates

Note: Latest edition of text books may be used.

	COURSE TITLE	**POMOLOGY-I**
1.	**Course Code**	HSC 303
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** on 2 hrs./week)**(Practical)** hrs./week)	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based (1 credit = 30 hours of practical based on 4 (1 Semester = 15 weeks**)** 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]

5. Max. Marks, Time and Instructions

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks

(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions.

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.

i) All questions carry equal marks.

ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Horticulture Inspector
7.	**Course Objectives**	To make the students familiar with basics of fruit production.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart basic knowledge about the importance and management of temperate fruits grown in India. CO 2. Study of commercial varieties of regional, national and international importance, ecophysiological requirements, recent trends in propagation, CO 3. Knowledge of industrial and export potential, Agri Export Zones (AEZ
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10 Course Content

UNIT– I

Principles of breeding of horticultural crops improvement of fruits. Importance of fruits and vegetable in human nutrition. Area, production and Contribution of Horticulture fruit crops in National Economy and exports.
Programmes of development - National Horticulture Mission.

UNIT– II

Growth and Development physiology of flowering, fruit set and development, parthenocarpy and seedlessness.Maturity and rip

UNIT– III

Protected cultivation training, pruning and canopy management. Biotic and abiotic stress breeding, biotechnological tools for breeding for fruits. Micropropogation, Meristem culture, ovule culture, in vitro pollination.

UNIT-IV

Production of' bio-agents and bio-fertilizer and green house management.
Importance of post harvest handling in horticulture crops.

Practical work:

1. Visit to identify temperate, tropical and sub tropical fruit plants.
2. Visit to fruit processing units.
3. Pruning practices in horticultural crops.
4. Maturity indices for various horticultural crops, handling, storage and packing techniques.
5. Visit to private orchards to identify different layout and cultural practices.

Suggested Resources for Reading

1. Adams, C.R. and M. P. Early. 2004. Principles of Horticulture. Butterworth – Heinemam, Oxford University Press.
2. Chadha, K.L. 2001, Handbook of Horticulture, ICAR, New Delhi.
3. Chattopadhyaya, P.K.2001. A Text Book on Pomology (Fundamentals of Fruit Growing) Kalyani Publication, New Delhi
4. Christopher, E.P. 2001. Introductory Horticulture, Biotech Books, New Delhi
5. Edmond, J.B. T.L. Senn, F.S. Andrews and P.G.Halfacre, 1975. Fundamentals of Horticulture, Tata MC. Graw Hill Publishing Co.New Delhi
6. George Acquaah, 2002, Horticulture-principles and practices. Prentice-Half of India pvt. Ltd., New Delhi.
7. Hartman, H.T. and Kester, D.E. 1986. Plant Propagation – Principles and Practices – Prentice Hall of India Ltd., New Delhi.
8. Jitendra Singh. 2006. Basic Horticulture. Kalyani Publishers, New Delhi.
9. Rajan, S. and B.L. Markose. 2007. Propagation of Horticultural Crops. New India Publishing, New Delhi.
10. Singh, N.P. 2005. Basic Concepts of Fruit Science. International Book Distributing Co., Lucknow

Teachers and students to surf concerned/internet sites including Wikipedia for subject/topic Updates

Note: Latest edition of text books may be used.

	COURSE TITLE	INTRODUCTORY SOIL SCIENCES
1.	**Course Code**	**HSC 401**
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks**)** 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5. Max. Marks, Time and Instructions		

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.

i) All questions carry equal marks.
ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Soil Testing Technicians in laboratory
7.	**Course Objectives**	To teach introductory knowledge of weathering and study of properties of soil.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1: To be able about physical and chemical properties of soil and their effect on plant's health. CO2: To aware the students about causes, effects and remedies to prevention and mitigation of soil pollution. CO3: Knowledge about soil forming rocks and minerals, their weathering and soil forming processes and climatic factors affect them.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10 **Course Content**

UNIT– I

Definition & components of soil
Definition, occurrence & classification of soil forming rocks
Soil Profile & various Soil Horizons

UNIT– II

Soil texture, soil structure, densities and porosity and factors affecting
Importance of Soil structure
Soil Organic Matter origin, its nature and composition

UNIT– III

Decomposition of various organic compounds and Factors affecting decomposition, decompositions of water soluble and water insoluble organic matters
Advantages and disadvantages of organic matter

UNIT-IV

Plant growth concept and factors affecting plant growth.
Plant nutrition: Essential elements, classification & general function of plant nutrients, forms of nutrient elements absorbed by plants
Process of nutrient uptake Soils of India.

Practical work:

1. Examination of soil profile in the field
2. Soil sampling and use of augers
3. Study of soil texture and soil structure of a given sample
4. Calculations of soil densities and porosity of soil
5. Soil survey of a farm

Suggested Resources for Reading

1. Brady, N.C. and Weil, R.R., The Nature and Properties of Soil: 13th edn., Pearson education Pte. Ltd. New Delhi, 2002
2. Oswal, M.C., Soil Physics, Oxford & IBH publishing Co. Pvt. Ltd. New Delhi, 1994
3. Biswas, T.D., and Mukherjee, S.K., Text book of soil science, Tata McGraw Hill Publishing Co. Ltd., New Delhi,1997

Teachers and students to surf concerned/internet sites including Wikipedia for subject/topic updates

Note: Latest edition of text books may be used.

	COURSE TITLE	INTRODUCTORY SEED TECHNOLOGY
1.	**Course Code**	HSC 402
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks**)** 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5	**Max. Marks, Time and ıstructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Supervisor in seed industries
7.	**Course Objectives**	To learn basic principles in seed production industry and release of varieties, their multiplications.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO-1: Start a seed production program for fill full the requirement of quality seed in market and increase the income. CO-2: Storage the pure variety seed to avoid the availability crises of pure variety seed due to adverse environmental conditions. CO-3: To supply disease-free seed in the market to promote the environmentally friendly cultivation of crops. CO-4: To increase the farm income by producing high-yielding, disease-free quality seed and decreasing the cost of cultivation.

9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc

10 Course Content

UNIT– I

Definition, uses, morphology and types of seed
Seed health, factors affecting seed health, production of healthy seed
Seed viability and germination of seed, factor affecting germination
Seed vigour and factor affecting seed vigour

UNIT– II

Seed dormancy, Seed deterioration, its causes and prevention of seed deterioration
Seed purity and purity test components

UNIT– III

Stages of production of improved seeds, Seed production organizations in India, Seed processing and seed treatments ,Packaging of seeds, Seed storage and factors affecting seed longevity in storage

UNIT-IV

Multiplication of seed (breeder seed, foundation seed, registered seed and certified seed)
Seed certification and its objectives

Practical work:

1. Identification of different type of seeds
2. Study of method of seed health tests: - inspection of dry seeds, washing test and Blotter method
3. Seed purity test
4. Study of Seed germination test
5. Analysis Seed viability test

Suggested Resources for Reading

1. Vishunavat, K., Seed health testing, principal and protocol, Kalyani Publishers, New Delhi,2009
2. Sen Subir, and Ghosh N, Seed Science and technology, Kalyani Publishers, New Delhi, 2009
3. Singh, B.D., Plant Breeding Principles and Methods, Kalyani Publishers, New Delhi, 2009

Teachers and students to surf concerned/internet sites including Wikipedia for subject/topic Updates

Note: Latest edition of text books may be used.

	COURSE TITLE	**POMOLOGY -II**
1.	**Course Code**	HSC 403
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks**)** 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:
The syllabus of this paper has been divided into FOUR units.

i) All questions carry equal marks.
ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Horticulture Supervisor
7	**Course Objective**	To make the students familiar with basics of fruit production, green house management and post harvest handling in horticulture crops
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart basic knowledge about the importance and management of temperate fruits grown in India. CO 2. Study of commercial varieties of regional, national, and international importance, eco-physiological requirements, recent trends in propagation, CO 3. Knowledge of the production of bio-agents, bio-fertilizer, and greenhouse management, along with post-harvest handling in horticulture crops

9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10 Course Content

UNIT– I

Principles of breeding of horticultural crops improvement of fruits. Importance of fruits and vegetable in human nutrition. Area, production and Contribution of Horticulture fruit crops in National Economy and exports.
Programmes of development - National Horticulture Mission.

UNIT– II

Growth and Development physiology of flowering, fruit set and development, parthenocarpy and seedlessness.Maturity andripening plant growth regulators and their role. Physiological basis and morphology of flowering and fruiting (berry, stone, pome, nut).

UNIT– III

Protected cultivation training, pruning and canopy management. Biotic and abiotic stress breeding, biotechnological tools for breeding for fruits. Micropropogation, Meristem culture, ovule culture, in vitro pollination.

UNIT-IV

Production of' bio-agents and bio-fertilizer and green house management.
Importance of post harvest handling in horticulture crops.

Practical work:

1. Visit to identify temperate, tropical and sub tropical fruit plants.
2. Visit to fruit processing units.
3. Pruning practices in horticultural crops.
4. Maturity indices for various horticultural crops, handling, storage and packing techniques.
5. Visit to private orchards to identify different layout and cultural practices.

Suggested Resources for Reading

1. Adams, C.R. and M. P. Early. 2004. Principles of Horticulture. Butterworth – Heinemam, Oxford University Press.
2. Chadha, K.L. 2001, Handbook of Horticulture, ICAR, New Delhi.
3. Chattopadhyaya, P.K.2001. A Text Book on Pomology (Fundamentals of Fruit Growing) Kalyani Publication, New Delhi
4. Christopher, E.P. 2001. Introductory Horticulture, Biotech Books, New Delhi
5. Edmond, J.B. T.L. Senn, F.S. Andrews and P.G.Halfacre, 1975. Fundamentals of Horticulture, Tata MC. Graw Hill Publishing Co.New Delhi
6. George Acquaah, 2002, Horticulture-principles and practices. Prentice-Half of India pvt. Ltd., New Delhi.
7. Hartman, H.T. and Kester, D.E. 1986. Plant Propagation – Principles and Practices – Prentice Hall of India Ltd., New Delhi.
8. Jitendra Singh. 2006. Basic Horticulture. Kalyani Publishers, New Delhi.
9. Rajan, S. and B.L. Markose. 2007. Propagation of Horticultural Crops. New India Publishing, New Delhi.
10. Singh, N.P. 2005. Basic Concepts of Fruit Science. International Book Distributing Co., Lucknow

Teachers and students to surf concerned/internet sites including Wikipedia for subject/topic Updates

Note: Latest edition of text books may be used.

	COURSE TITLE	**DISEASES AND MANAGEMENT OF HORTICULTURE CROPS**
1.	**Course Code**	HSC 404
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks**)** 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5. Max. Marks, Time and Instructions		

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Supervisor in plant protection clinics
7.	**Course Objectives**	To acquaint with diseases of horticulture crops and their management.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO-1.Student will know the common pathogens of different diseases. CO-2. Student acquire the knowledge about etiology, and symptoms of these diseases which helps in diagnosis of the diseases of horticultural crops CO-3. By knowing means of dispersal of these diseases suitable management methods can be applied. CO-4. Eco-friendly and economically suitable management practices may be adopted.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10 **Course Content**

UNIT– I

Symptoms, host range their management and control of different diseases of fruits like apple, mango, citrus, papaya, sapota, grapes and guava.

UNIT– II

Symptoms, host range their management and control of different diseases of fruits like pear, peach, plum, apricot, ber, and loquat.

UNIT– III

Symptoms, host range their management and control of different diseases of vegetables like brinjal, melon, pumpkin, sweet potato, cabbage, potato, onion and peas.

UNIT-IV

Symptoms, host range their management and control of different diseases of vegetables like chilies and Turmeric.

Practical work:

- Detailed study of symptoms and their management of horticultural and vegetables diseases of the above mentioned crops
- Collection and dry preservation of diseased specimens of important crops

Suggested Resources for Reading

1. Gupta VK &.Sharma SK. 2000. Diseases of Fruit Crops. Kalyani Publ., New Delhi
2. Pathak VN. 1980. Diseases of Fruit Crops. Oxford & IBH, New Delhi
3. Singh RS. 2000. Diseases of Fruit Crops. Oxford & IBH, New Delhi
4. Walker JC. 2004. Diseases of Vegetable Crops. TTPP, India
5. Package of Practices for Vegetable Crops, P.A.U. Publications Ludhiana, Corresponding year

Teachers and students to surf concerned/internet sites including Wikipedia for subject/topic updates

Note: Latest edition of text books may be used.